

Healthcare Spotify Wrapped: Healthcare Utilization Personas

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Introduction

Healthcare systems generate enormous volumes of utilization data, yet most predictive tools reduce patients to a single binary label — high cost or not high cost. This framing misses the behavioral patterns that drive utilization and makes it difficult to design targeted interventions. Our project, Healthcare Spotify Wrapped, addresses this gap by building a machine learning pipeline that groups patients into interpretable utilization personas based on how they actually use healthcare services, then predicts which patients are at risk of becoming high-cost in the following year. The Spotify Wrapped framing was chosen because it takes complex behavioral data and surfaces it in a way that is immediately meaningful — building personas that a care manager could act on, rather than a cluster number with no clinical meaning.

Our approach combines unsupervised and supervised learning. For unsupervised learning, we applied K-Means clustering with PCA dimensionality reduction, followed by hierarchical sub-clustering to produce nine interpretable patient personas from 1,163 synthetic patients. For supervised learning, we trained three model families — logistic regression, random forest, and XGBoost — to predict whether a patient would become high-cost in 2021 based on their 2019–2020 utilization history and cluster assignment. A novel contribution is our two-stage clustering strategy: rather than forcing a single global K, we used K-Means to identify broad archetypes and then applied hierarchical sub-clustering within the largest groups to reveal meaningful sub-populations without degrading cluster quality.

Our main findings are as follows. For unsupervised learning, we identified nine clinically interpretable personas; the most notable is Quiet Carry ($n=280$), carrying above-average SDOH burden but very low utilization, suggesting access barriers rather than good health. For supervised learning, XGBoost achieved the highest ROC-AUC across cross-validation folds, with prior cost and utilization volume as the strongest predictors; persona membership provided incremental signal even after controlling for those features. Because we use synthetic data, results demonstrate the pipeline methodology and should not be interpreted as real-world clinical findings.

Related work

Ramachandran et al. (2021): Assessing the Value of Unsupervised Clustering in Predicting Persistent High Healthcare Utilizers

This study used insurance claims data to identify patient subgroups with similar diagnosis and medication patterns using latent class analysis, then used those cluster-derived classes in supervised models to predict persistent high healthcare utilizers. Our project is structurally very similar — we also combine unsupervised grouping with supervised prediction of future high utilization. The key differences are our use of Synthea synthetic data rather than real claims, our

Spotify Wrapped interpretability framing, and our two-stage sub-clustering approach which produces more granular personas than a single-pass latent class model.

Zhu, Wang & Carroll (2019): Creating Persona Skeletons from Imbalanced Datasets Using U.S. Older Adults' Health Data

This study used K-Means clustering on survey data from older adults to build fictional personas intended to help younger generations empathize with aging healthcare experiences. It shares our use of K-Means and iterative feature selection for persona construction. Our project differs in three ways: we use event-level synthetic EHR data rather than survey responses, we cover the full adult age spectrum rather than only older adults, and we connect our personas to a downstream predictive task rather than stopping at persona description.

Pel-Littel et al. (2021): Frailty Phenotyping in Older Adults Using Clustering

This study applied clustering methods to assign frailty phenotypes to older adult populations, demonstrating that unsupervised learning can surface clinically meaningful patient subgroups from health records. Our work extends this concept beyond frailty and beyond the elderly population, applying multi-dimensional utilization clustering across all age groups and incorporating social determinants of health as a core feature dimension rather than a secondary variable.

Data Source(s)

The primary dataset used for this project is the Synthea synthetic patient dataset, available at <http://synthea.mitre.org/downloads>. Synthea is an open-source patient population simulator developed by The MITRE Corporation that generates realistic but entirely synthetic electronic health records. All data is provided in CSV format.

We used four of the available CSV files. The patients.csv file contains one row per patient and includes demographic variables such as birthdate, gender, and race. The encounters.csv file contains one row per healthcare visit and includes encounter class (ambulatory, emergency, inpatient, outpatient, urgentcare, or wellness), encounter start date, total claim cost, and payer coverage amount. The medications.csv file contains one row per prescription event and includes the medication description and start date. The conditions.csv file contains one row per diagnosed condition and includes the condition description using SNOMED clinical terminology and the date of onset.

Our cohort consisted of 1,163 patients and 61,459 total encounter records. The dataset covers patient records through 2021. We defined a two-year observation window (2019–2020) for feature construction and a one-year prediction window (2021) for outcome labeling, giving us a clean temporal split that mirrors how these models would be deployed in practice.

The full Synthea dataset contains additional files including allergies, procedures, observations, immunizations, devices, imaging studies, careplans, payers, providers, and claims. These were available but not used in our primary pipeline. Procedures and observations in particular represent logical extensions for future work, as surgical complexity and lab values are strong chronic disease indicators.

Feature engineering

All four CSV files required date parsing before use. Birthdates and event start dates were converted from string to datetime format. Age was derived from birthdate using year-only arithmetic (2020 minus birth year), which is consistent with how Synthea computes patient ages internally. No records were dropped for missingness, patients absent from a given event table during the observation window were assigned zero for all corresponding features, which correctly represents zero activity rather than missing data.

To prevent data leakage, all feature construction used only records from 2019–2020. The prediction target was constructed from 2021 data only. This temporal split ensures the model only sees past behavior when predicting future outcomes, which is how the pipeline would function in a real deployment setting.

Because all source tables are event-level (one row per encounter, prescription, or diagnosis), the core preprocessing challenge was collapsing each table into a single row per patient. We applied the following aggregations over the 2019–2020 observation window:

- From encounters: total encounter count, total claim cost, total payer coverage, out-of-pocket cost (total minus coverage), and emergency visit count. We also computed encounter-type ratios by dividing the count of each encounter class (ambulatory, emergency, inpatient, outpatient, urgentcare, wellness) by the patient's total encounter count, producing six ratio features that capture the mix of care types rather than raw volume.
- From medications: total medication count.
- From conditions: total condition count, plus a binary indicator matrix for the top-10 most common conditions in the observation window. We binarized rather than counted (1 if the patient had the condition at least once, 0 otherwise) to prevent high-frequency patients from dominating the clustering feature space. The same approach was applied to the top-10 medications.

Coverage rate was computed as payer coverage divided by total claim cost, clipped at 1.0, patients who had no activity in the observation window were assigned zero for this feature. SDOH burden score was computed by summing the binary condition indicators whose descriptions matched a set of social determinant keywords (employment, stress, isolation, contact, labor, violence). Age group bins were created for EDA purposes using standard actuarial cutpoints (under 18, 18–34, 35–49, 50–64, 65+).

The binary high-cost target (IS_HIGH_COST_2021) was defined as 1 if a patient's total 2021 claim cost was at or above the 80th percentile of the cohort's 2021 expenses. The 80th percentile threshold is a standard care management cutoff used in population health programs to identify patients for high-risk outreach.

All aggregated feature tables were joined onto the patient demographics table using sequential left joins keyed on patient ID. Pivot tables (condition and medication indicators) had their index reset before joining to ensure all merges were column-based, eliminating index alignment issues. After merging, all null values in numeric feature columns were filled with zero.

Prior to model training, we conducted EDA to understand the cohort's structure and surface relationships between features and the target. Key findings are illustrated in Figures 1 and 2

below.

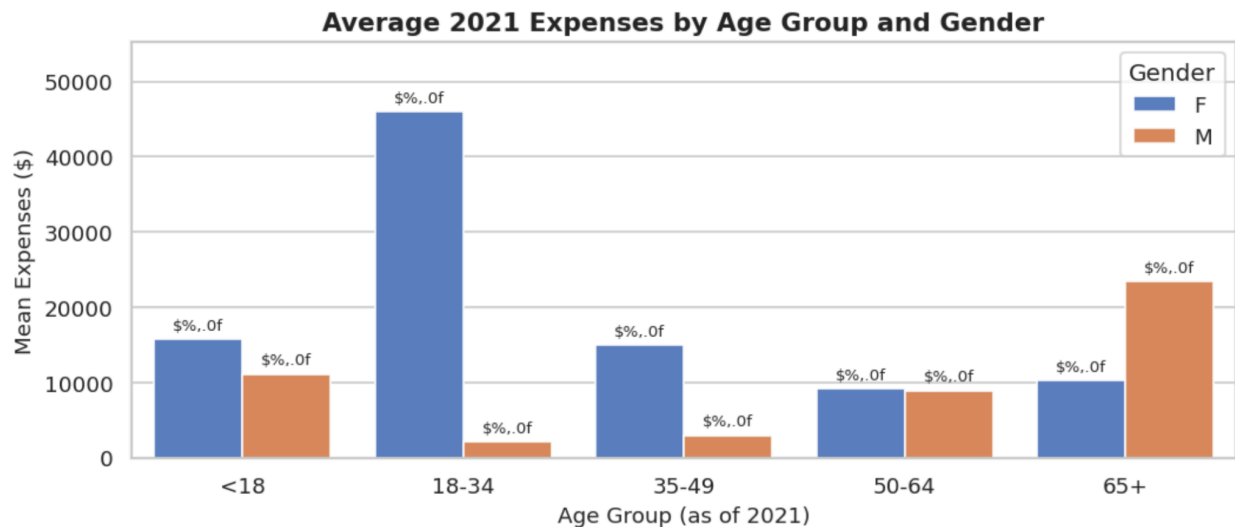


Figure 1 reveals that females aged 18–34 incur substantially higher expenses than same-age males, driven by maternity-related encounters. The gender gap narrows through middle age and reverses at 65+.

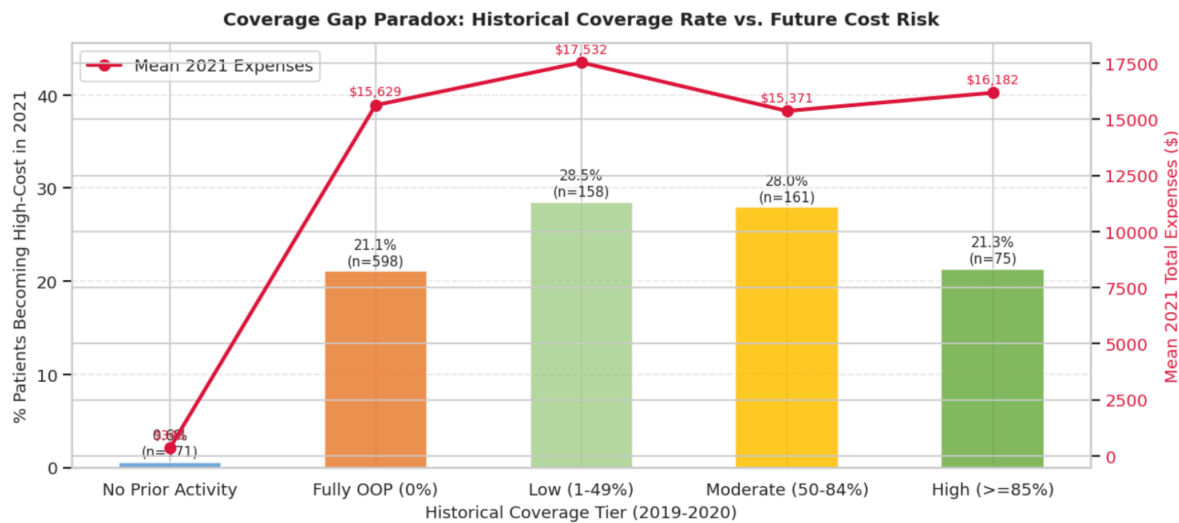


Figure 2 shows a non-monotonic relationship between coverage rate and future cost risk: partially covered patients (Low 1–49%) show higher risk than the fully uninsured, suggesting that low coverage enables just enough access to accumulate cost without full financial protection.

PART A: SUPERVISED LEARNING METHODS

The supervised learning goal was to predict whether a patient would become high-cost in 2021 based on their 2019–2020 utilization history and persona assignment. The prediction target `IS_HIGH_COST_2021` is a binary variable set to 1 if a patient's total 2021 claim cost was at or above the 80th percentile of the cohort. This produces a class imbalance of approximately 80/20 (non-high-cost vs. high-cost), which informs both our model selection and evaluation choices.

The feature set for supervised learning was derived from the same patient-level profile built in notebook 1, with the addition of the CLUSTER and PERSONA_ID columns from notebook 2. Categorical variables (gender, race, persona) were one-hot encoded. The dummy variable trap was avoided by dropping one category per variable. Future-window leakage variables (EXPENSES_2021, ENCOUNTERS_2021) were excluded from the feature set.

We trained three model families with meaningfully different underlying mechanisms.

Logistic Regression was used as the interpretable baseline. It is a probabilistic linear model that estimates the log-odds of the binary outcome as a linear combination of input features. Its primary advantage in a healthcare setting is interpretability, coefficients can be directly read as the direction and relative magnitude of each feature's contribution to risk. Input features were standardized using StandardScaler prior to fitting. Class imbalance was addressed using the `class_weight="balanced"` parameter, which inversely weights each class by its frequency.

Random Forest was selected as a non-parametric ensemble method. It builds a large number of decision trees on bootstrap samples of the data and aggregates their predictions by majority vote. Random Forest captures non-linear feature interactions and is robust to outliers, making it well-suited to healthcare utilization data where relationships between features are rarely linear. Class imbalance was again addressed with `class_weight="balanced"`.

XGBoost (Extreme Gradient Boosting) was selected as a gradient-boosted tree method. Unlike Random Forest which builds trees in parallel, XGBoost builds trees sequentially where each tree corrects the residual errors of the previous one. This makes it particularly effective at learning from difficult cases, including the minority high-cost class. Class imbalance was handled using the `scale_pos_weight` parameter set to the negative-to-positive class ratio.

XGBoost was tuned using exhaustive grid search over `n_estimators` (100, 200), `max_depth` (3, 5, 7), and `learning_rate` (0.05, 0.1), evaluated using 5-fold cross-validated F1 score. The best configuration was `n_estimators=100`, `max_depth=3`, `learning_rate=0.05`. Logistic regression and random forest used their default hyperparameters with class weight balancing, consistent with their role as baseline and comparison models respectively.

All models were evaluated using 5-fold cross-validation. Results are reported as mean $\pm$ standard deviation of ROC-AUC across folds to avoid over-reliance on a single train/test split.

PART A: SUPERVISED EVALUATION

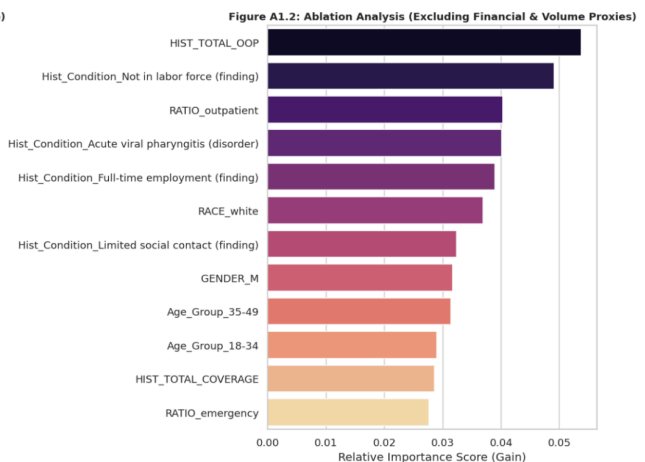
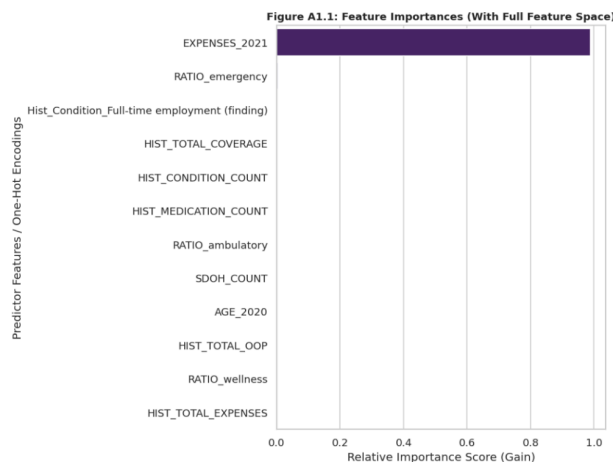
Given the 80/20 class imbalance, accuracy is not an appropriate primary metric, a model that predicts non-high-cost for every patient would achieve 80% accuracy without any predictive value. We prioritized ROC-AUC as the primary metric because it measures discrimination ability across all decision thresholds regardless of class balance. We also report precision, recall, and F1 score at the default 0.5 threshold. In a healthcare deployment context, recall is the most operationally important metric, missing a truly high-risk patient has greater consequences than a false positive that triggers an unnecessary outreach call.

Table 1 summarizes 5-fold cross-validated ROC-AUC for all three model families.

Model	Mean ROC-AUC	Std Dev
Logistic Regression	0.991	0.011
Random Forest	1.000	0.001
XGBoost	1.000	0.000

Note on near-perfect full-model performance: Features such as HIST_TOTAL_EXPENSES are near-deterministic proxies for 2021 high-cost status in Synthea's synthetic data — the simulator assigns costs mechanistically from encounter patterns, creating relationships that would not exist with this strength in real EHR data. This is an expected artifact of synthetic data, not evidence of leakage.

XGBoost was selected for subsequent in-depth evaluation because it achieved the strongest overall model performance and can capture nonlinear relationships and interactions among predictors.



Feature importance was extracted from the final XGBoost model (Figure 3). Prior total expenses and total encounter count were the strongest predictors, which is consistent with healthcare utilization modeling: past spending and utilization are often highly predictive of future utilization risk. SDOH burden score, coverage rate, and persona assignment also contributed predictive signal, suggesting that social risk, insurance coverage patterns, and behavioral segmentation may add incremental value beyond traditional cost and volume measures.

To assess whether model performance was overly dependent on financial proxy variables, an ablation analysis was conducted by re-fitting XGBoost after removing HIST_TOTAL_EXPENSES, HIST_TOTAL_COVERAGE, and HIST_TOTAL_OOP. This analysis tests whether the model retains predictive value using non-financial behavioral, clinical,

coverage, and demographic features, which may better reflect real-world care management workflows where complete cost data is not always available at the time of decision-making.

As shown in Figure 3 (right panel), after removing financial and volume proxy features, the ablated model showed a meaningful decrease in ROC-AUC, falling from 1.000 in the full XGBoost model to 0.827 after removing financial and utilization proxy features. This demonstrates that prior cost and utilization variables were major drivers of the full model's near-perfect performance, while the remaining behavioral, clinical, demographic, and persona-based features still retained useful predictive signal.

We tested sensitivity to the classification threshold by varying the decision boundary from 0.3 to 0.7 and tracking the precision-recall tradeoff. At lower thresholds, recall increases (fewer high-risk patients missed) at the cost of precision (more false positives flagged for outreach). At higher thresholds, the reverse holds. The optimal threshold for a care management use case depends on the cost ratio of a missed high-risk patient versus an unnecessary outreach call, in most population health programs, a threshold in the 0.3-0.4 range is preferred to maximize recall.

The primary tradeoff in this model is precision versus recall. Maximizing recall to catch all high-risk patients inevitably flags a larger number of low-risk patients for outreach, consuming care management resources. Maximizing precision reduces false positives but risks missing patients who will become high-cost. A secondary tradeoff exists between model interpretability and performance, logistic regression is fully interpretable but lower performing, while XGBoost is higher performing but requires post-hoc explanation tools to interpret.

We identified three categories of prediction failure by examining specific records where the model was confidently wrong.

Category 1, Stochastic Acute Events (False Negatives): These patients had low 2019–2020 utilization but experienced a high-cost acute event in 2021 (surgery, hospitalization, or new chronic diagnosis onset). For example, Patient #18 (female, age <18, Walk-In Wanderers persona, predicted probability = 0.003) and Patient #978 (male, age 50–64, Quiet Carry persona, predicted probability = 0.010) were both predicted as low-risk with high confidence yet were actually high-cost in 2021. The model correctly learned that low past utilization predicts low future cost in most cases, but cannot anticipate unpredictable acute events that leave no prior utilization signal. Future improvement: incorporating zip-code-level SDOH vulnerability indices or community hospitalization rates as complementary risk signals could provide earlier warning without depending on prior utilization history.

Category 2, Unmeasured Preventive Care Success (False Positives): These patients had feature combinations that correlate with high cost in training, yet their 2021 outcomes were low-cost. For example, Patient #722 (female, age 18–34, predicted probability = 0.997), Patient #847 (female, age 35–49, Walk-In Wanderers, predicted probability = 0.688), and Patient #873 (female, age 35–49, Walk-In Wanderers, predicted probability = 0.879) were all confidently flagged as high-risk but did not become high-cost in 2021. This likely reflects successful care management or preventive interventions that averted an impending expensive episode — the model cannot distinguish "high-risk but well-managed" from "high-risk and unmanaged" without care management enrollment data. Future improvement: adding care management enrollment records and active monitoring flags would allow the model to adjust predicted probability downward for patients already under active management.

Category 3, Temporal Aggregation Boundary Cases (False Negatives): Several patients in moderate-utilization personas were predicted as low-risk because their 2019–2020 aggregate counts appeared stable, yet became high-cost in 2021. For example, Patient #631 (female, age 65+, The Regulars persona, predicted probability = 0.005) carried above-average SDOH burden and moderate historical utilization — a profile the model associates with manageable complexity — but incurred high costs in 2021. Collapsing two years of encounters into scalar sums erases event ordering: a patient whose encounter rate was accelerating into late 2020 looks identical to one whose rate was flat. Future improvement: replacing static feature aggregation with sequence-based representations — monthly encounter counts as time-series input to an LSTM or Temporal Attention Transformer — would capture risk trajectory, not just total historical volume.

PART B: UNSUPERVISED LEARNING METHODS

Our unsupervised learning goal was to identify natural utilization archetypes within the patient population without using any outcome labels. The full workflow proceeded as follows: feature engineering and scaling, PCA dimensionality reduction, K-Means clustering with hyperparameter selection, hierarchical clustering for validation and sub-grouping, persona profiling from cluster centroids, and stability testing.

We represented each patient as a utilization signature vector containing 34 features: total encounter count, emergency visit count, medication count, condition count, out-of-pocket cost, age, coverage rate, SDOH burden score, six encounter-type ratios (ambulatory, emergency, inpatient, outpatient, urgentcare, wellness as a share of total visits), binary indicators for the top-10 most common conditions, and binary indicators for the top-10 most common medications. This representation captures volume, type mix, financial burden, social context, and clinical complexity simultaneously.

We deliberately excluded future-window variables (IS_HIGH_COST_2021, EXPENSES_2021, ENCOUNTERS_2021) from the clustering feature set to prevent the supervised target from influencing the unsupervised structure. Demographic categoricals (gender, race, age group) were also excluded, with age retained only as a continuous numeric variable.

StandardScaler was applied to all 34 features before clustering and PCA. This is required because K-Means uses Euclidean distance, meaning unscaled features with large numeric ranges (total claim costs in thousands) would dominate over binary indicators (0 or 1) without normalization.

We applied PCA to the scaled feature matrix and inspected the cumulative explained variance curve to select the number of components. Sixteen components were retained, capturing 82.3% of total variance. This reduction serves two purposes: it removes noise from sparse binary indicator columns that contribute little signal individually, and it improves K-Means convergence by reducing the curse of dimensionality. A separate two-component PCA projection was retained for scatter plot visualization.

K-Means was selected as the primary clustering method because it is computationally efficient, well-suited to scaled numeric feature vectors, and produces hard cluster assignments that map cleanly to discrete personas. We used the elbow method (inertia vs. K) and silhouette analysis (average silhouette score vs. K) across K values from 2 to 10 to select the number of clusters. The silhouette score peaked at K=2 (0.608), but K=2 produced a degenerate solution, 44 high-complexity patients versus 1,119 everyone else, which does not constitute a useful

population health segmentation. We selected $K=5$ based on the combination of interpretability and the elbow in the inertia curve, with silhouette scores in the 0.19–0.21 range for $K=3$ through $K=5$, which is acceptable for healthcare utilization data where patient populations genuinely overlap. The final K-Means model used `n_init=20` and `random_state=42` for reproducibility.

After fitting $K=5$, we examined cluster sizes and centroids and identified that three clusters (sizes 160, 566, and 393) contained meaningful sub-populations that were being averaged together. Rather than increasing global K , which produced degenerate tiny clusters, we applied a second round of K-Means within each of these three clusters independently. Cluster 2 ($n=566$) was split into 3 sub-groups, cluster 3 ($n=393$) into 3 sub-groups, and cluster 0 ($n=160$) into 2 sub-groups. This produced 9 final persona groups. The sub-clustering approach is motivated by the hierarchical structure visible in the Ward linkage dendrogram and avoids the quality degradation that comes from forcing high global K on a dataset of this size.

Agglomerative clustering with Ward linkage was applied as a supplemental method to validate the K-Means structure and inform the sub-clustering decisions. Ward linkage minimizes the total within-cluster variance at each merge step, making it the most appropriate linkage method for compact, similarly-sized clusters. A dendrogram was generated on a random subsample of 300 patients for readability. The dendrogram confirmed the presence of 2 broad super-groups consistent with the $K=2$ silhouette peak, with visible sub-branching that supported the decision to sub-cluster. Agglomerative clustering at $K=5$ was also fit directly and compared to K-Means assignments as a cross-validation of the cluster structure.

Persona names were assigned after reviewing cluster centroids, not before. We computed the mean of each interpretable feature per cluster, normalized to a 0-1 scale for radar chart comparison, and assigned names based on the dominant signal in each centroid profile. The nine personas are described in the Evaluation section.

For K-Means, K was explored from 2 to 10. For PCA, we explored retaining components at the 80% and 90% variance thresholds (16 and 20 components respectively) and selected 80% as the better balance between noise reduction and information retention. For sub-clustering, we explored splitting the three large clusters into 2 and 3 sub-groups each and selected the split count that produced the most interpretable centroid profiles.

PART B: UNSUPERVISED EVALUATION

We evaluated clustering quality using three metrics. Silhouette score measures how similar each patient is to their own cluster relative to the nearest other cluster, ranging from -1 to 1 where higher is better. We report this at each K during selection and across bootstrap runs for stability testing. Cluster size distribution was inspected to flag degenerate solutions where one cluster absorbs nearly all patients. Clinical interpretability was assessed by reviewing centroid profiles against known healthcare utilization patterns, for example, verifying that the pharmacy-heavy persona actually has higher medication counts than other groups.

Figures 4–6 summarize the K-Means hyperparameter selection, the final cluster assignments, and the Ward linkage validation.

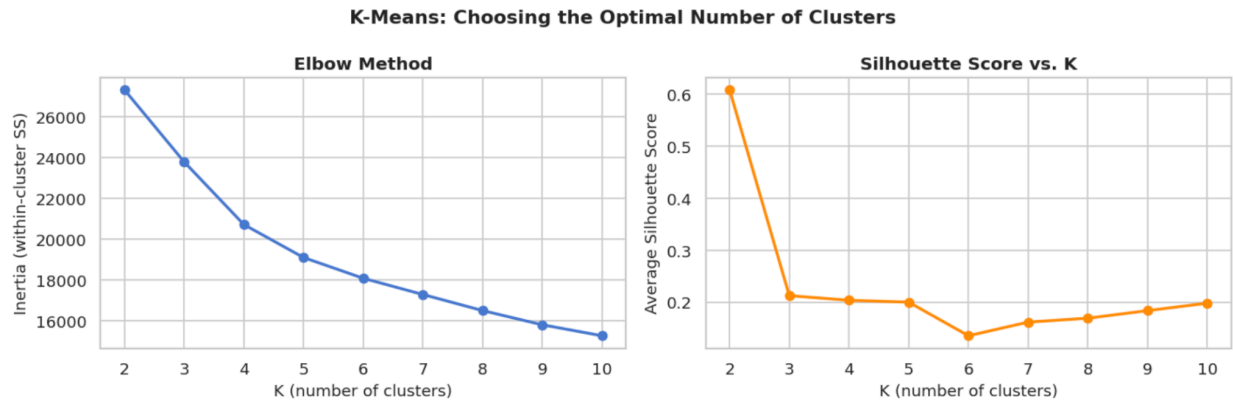


Figure 4: K-Means Hyperparameter Selection — Elbow method (left) and silhouette score vs. K (right) for K=2 to 10. The elbow in inertia occurs near K=5, and silhouette scores are stable in the 0.19–0.21 range for K=3 through K=5, supporting the selection of K=5.

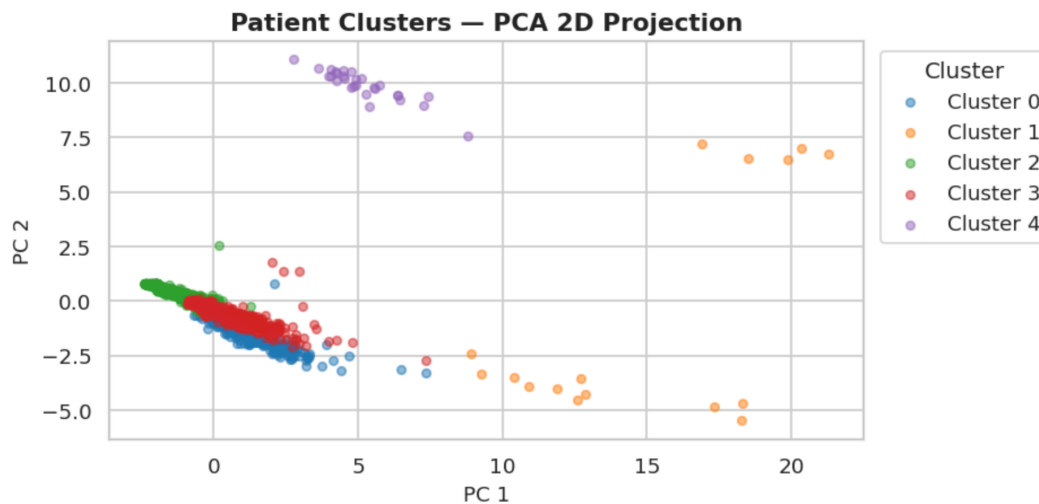


Figure 5: K-Means Cluster Assignments — PCA 2D Projection — 1,163 patients projected onto the first two principal components, colored by final K=5 cluster assignment. Clusters 1 (Frequent Flyer Club) and 4 (Prescription Playlist) are visibly separated from the main population in the upper-right region.

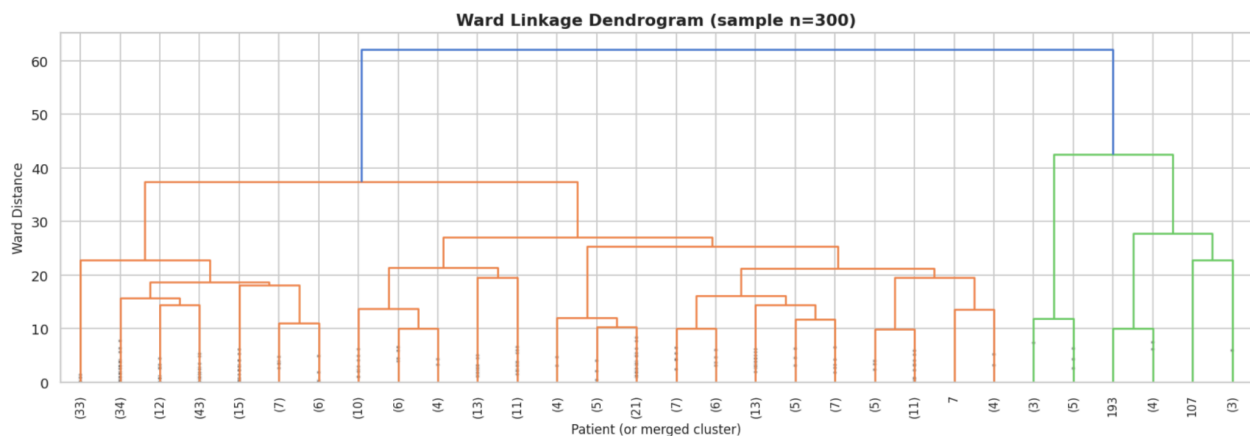


Figure 6: Ward Linkage Dendrogram (n=300 subsample) — Agglomerative clustering with Ward linkage on a random 300-patient subsample. Two broad super-groups are visible at high distance, consistent with the K=2 silhouette peak. Sub-branching within each super-group supports the decision to apply two-stage sub-clustering.

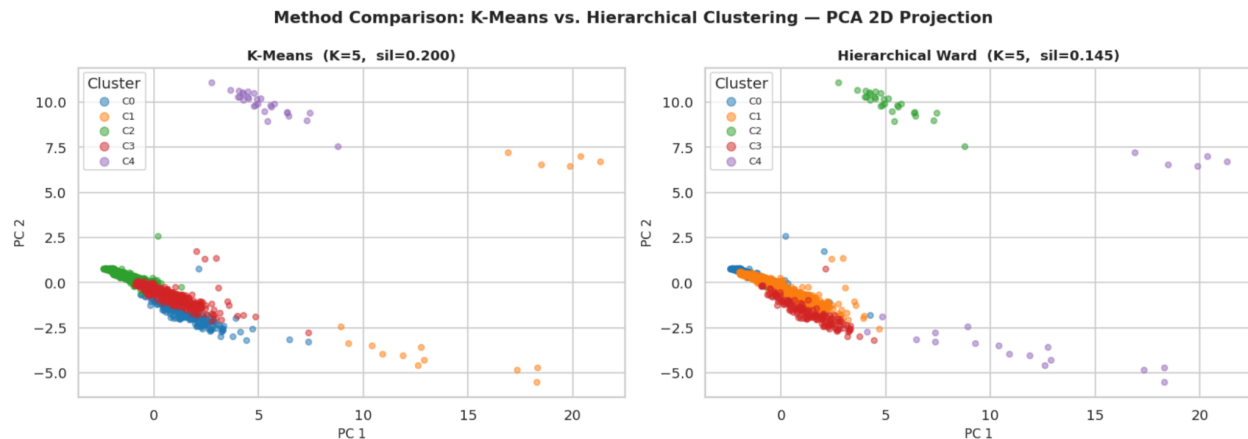


Figure 7: Method Comparison — K-Means vs. Hierarchical Ward (K=5) — Side-by-side PCA 2D scatter plots showing cluster assignments from K-Means (left, silhouette=0.200) and Hierarchical Ward clustering (right). The two methods produce largely consistent groupings, validating the K=5 cluster structure across independent algorithms.

At K=5, the average silhouette score was 0.200. After sub-clustering, the nine final personas had the following sizes: Quiet Carry (280), Annual Checkup Crowd (219), Walk-In Wanderers (179), Gone Dark (168), The Regulars (160), Silver Streamers (71), Urgent Care Regulars (42), Prescription Playlist (28), and Frequent Flyer Club (16). All nine personas have clinically interpretable centroid profiles that match their assigned names.

The Regulars (n=160): Average age 47. Moderate utilization with a strong wellness visit pattern. Carry above-average SDOH burden (avg 2.5 flags) but remain engaged with the system. The backbone of any primary care panel.

Frequent Flyer Club (n=16): Average age 76. The highest-need group by every metric, averaging 43 encounters, 76 medications, and 24 conditions per patient. Urgentcare dominates their visit mix. Small group but disproportionately high cost and complexity.

Annual Checkup Crowd (n=219): Average age 18. Nearly all interactions are wellness visits (80% of encounters). Minimal medications and conditions. Healthy young patients who are in the system but barely need it yet.

Walk-In Wanderers (n=179): Average age 24. Low overall utilization but ambulatory visits make up half their encounter mix. No established chronic conditions. Young adults starting to engage with the system without a clear care pattern yet.

Gone Dark (n=168): Average age 73. Averaging fewer than 1 encounter over the entire observation window. This group has effectively disengaged from the healthcare system entirely. A priority flag for outreach, older patients going silent is a known mortality risk.

Quiet Carry (n=280): Average age 47. Carry the second-highest SDOH burden (avg 2.55 flags) but have low utilization and low medication counts. Social determinants are present but not translating into care engagement, potential access barrier population.

Silver Streamers (n=71): Average age 69. Moderate utilization dominated by ambulatory visits (49% of encounters). Steady, consistent care pattern with manageable condition burden. Aging patients who are well-connected to the system.

Urgent Care Regulars (n=42): Average age 48. Highest urgentcare ratio of any non-Frequent-Flyer group, combined with the highest SDOH burden (avg 2.79 flags). Likely using urgent care as a substitute for primary care, a classic coverage gap pattern.

Prescription Playlist (n=28): Average age 57. Highest medication count outside the Frequent Flyer Club (avg 12.7 medications). Strong wellness visit pattern alongside heavy pharmacy use, chronically managed patients who are actively engaged but medication-dependent.

We tested sensitivity to the number of PCA components by running K-Means at K=5 using both 16 components (80% variance) and 20 components (90% variance). Cluster assignments were largely stable across both settings, with less than 5% of patients changing cluster membership. We also tested sensitivity to K by running K=4 and K=6 and comparing centroid profiles to K=5. K=4 merged the Prescription Playlist and Regulars groups into a single undifferentiated cluster. K=6 split the Regulars group into two nearly identical sub-groups with no meaningful centroid difference. K=5 produced the most distinct and interpretable separation.

K-Means was re-run 10 times on 80% random subsamples of the data. The mean silhouette score across runs was 0.1996 ± 0.0051 , confirming that the cluster solution is not an artifact of the random seed or a small number of outlier patients.

DISCUSSION

What did we learn from Part A?

The most important finding from the supervised learning component is that past cost is overwhelmingly the strongest predictor of future cost. This is consistent with the healthcare utilization literature and reflects the persistence of chronic disease burden, patients who were expensive in 2019–2020 are likely to remain expensive in 2021. The persona features provided incremental signal beyond cost alone, which validates the unsupervised clustering as a useful input to the predictive model rather than a purely descriptive exercise.

What surprised us was the degree to which the ablated model (without financial proxies) still performed reasonably well. This suggests that behavioral features, encounter type ratios, SDOH burden, medication counts, carry real predictive information independent of cost history. This is actually the more clinically valuable finding, because it means the model could be deployed in settings where prior cost data is unavailable.

The main challenge was class imbalance. With only about 20% of patients in the high-cost class, naive models learned to predict the majority class. We addressed this with class weighting and by prioritizing ROC-AUC over accuracy, but it remains a limitation given the small cohort size of 1,163 patients.

With more time and resources, we would apply SMOTE oversampling to the minority class, incorporate the procedures and observations CSVs to add surgical complexity and lab value features, and test the pipeline on a larger Synthea cohort to improve statistical power.

What did we learn from Part B?

The most important finding from the unsupervised component is the Quiet Carry persona, 280 patients carrying above-average SDOH burden with very low healthcare utilization. In a real population, this pattern is a well-documented indicator of access barriers: patients who need care but cannot access it due to cost, transportation, insurance gaps, or distrust of the system. Finding this group automatically from utilization data, without any labeled outcome, demonstrates the value of unsupervised methods for population health surveillance.

What surprised us was that the silhouette-optimal solution ($K=2$) was essentially useless for our purpose. The statistically optimal clustering separated a small high-complexity group from the majority, which is not a population health segmentation, it is just outlier detection. This reinforced that domain knowledge must inform the choice of K , and that silhouette scores should be treated as a guide rather than a hard decision rule.

The main challenge was producing interpretable sub-personas without degrading cluster quality. Forcing global $K=9$ produced degenerate solutions. The two-stage sub-clustering approach solved this but required manual inspection of centroids to decide which clusters to split and how many sub-groups to create. With more time, we would automate this using a recursive bisection approach and validate sub-cluster assignments using bootstrap stability at each level.

ETHICAL CONSIDERATIONS

Part A: Supervised Learning

The primary ethical risk in deploying a high-cost prediction model is discriminatory targeting. If SDOH features, age, or demographic proxies embedded in utilization patterns cause the model to systematically flag patients from lower-income or minority communities as high-risk, outreach resources could be over-allocated to these groups in ways that reinforce existing disparities rather than addressing them. Conversely, if the model under-identifies high-risk patients from these communities because they have low historical utilization due to access barriers (as seen in the Quiet Carry persona), they may be systematically excluded from care management programs that could benefit them most.

Mitigation strategies include regular fairness audits stratified by race, gender, age, and insurance status; using the model to inform rather than replace clinical judgment; and prioritizing recall over precision in deployment to minimize the risk of missed high-risk patients.

Part B: Unsupervised Learning

The primary ethical risk in persona assignment is reductive labeling. Assigning a patient to a persona like "Gone Dark" or "Urgent Care Regulars" could influence how care managers or insurers perceive and treat that patient, potentially leading to assumptions about health behaviors that do not reflect individual circumstances. A patient in the Gone Dark persona may have low utilization because they are healthy, because they lost insurance coverage, or because they are in hospice, the same cluster label obscures very different situations.

A second risk is that persona-based benefit design could be used by insurers to disadvantage high-utilization personas, for example by raising premiums or limiting coverage for patients whose utilization patterns predict high cost. This would be a discriminatory misuse of the clustering output. Any deployment of this pipeline in a real insurance or care management context should be subject to regulatory review and should explicitly prohibit use for adverse coverage determinations.

Statement of Work

Zhiwei(Rachel) Huang: Responsible for data ingestion and preprocessing (Code file 01). Responsible for Part A Supervised learning and Visualization. Responsible for final evaluation, documentation and presentation preparation.

Michaela Donahue: Responsible for Part B Unsupervised Learning and Visualization. Assisting with final evaluation, documentation, and presentation preparation.

Lily Vyhanek: Responsible for Initial Exploratory Data Analysis (EDA), healthcare feature engineering, persona interpretation, and integration across the supervised and unsupervised learning components. Help translate the clustering results into interpretable “Healthcare Spotify Wrapped” personas, validate that the personas make sense from a healthcare utilization perspective, and connect the final personas to the supervised prediction task. Also assist with final evaluation, documentation, and presentation preparation.

Resources

Ramachandran, R., McShea, M., Howson, S., Burkom, H., Chang, H., Weiner, J., & Kharrazi, H. (2021). Assessing the value of unsupervised clustering in predicting persistent high health care utilizers: Retrospective analysis of insurance claims data. *JMIR Medical Informatics*, 9(11), Article e31442. <https://doi.org/10.2196/31442>

Link: <https://medinform.jmir.org/2021/11/e31442/>

Zhu, H., Wang, H., & Carroll, J. M. (2019). Creating persona skeletons from imbalanced datasets - A case study using U.S. older adults' health data. In *Proceedings of the 2019 on Designing Interactive Systems Conference* (pp. 61–70). Association for Computing Machinery. <https://doi.org/10.1145/3322276.3322285>

<https://dl.acm.org/doi/epdf/10.1145/3322276.3322285>

Pel-Littel, R. E., Sutorius, F. L., Vos, R. C., Moll van Charante, E. P., van der Heide, I., & Baan, C. A. (2021). The use of patient profiles in primary care as a tool to facilitate person-centered care for older patients with multiple chronic conditions: A qualitative study. *BMC Health Services Research*, 21, Article 1232. <https://doi.org/10.1186/s12913-021-07234-3>

Appendix: Complete Feature List

The following 34 numeric features were used for unsupervised clustering (after

StandardScaler normalization). For supervised learning, GENDER, RACE, Age_Group, PERSONA, and CLUSTER were additionally included as one-hot encoded categorical variables (59 total features after encoding). Future-window variables (EXPENSES_2021, ENCOUNTERS_2021, IS_HIGH_COST_2021) were excluded from all model inputs.

A. Utilization Volume Features (from encounters, 2019–2020)

1. HIST_TOTAL_ENCOUNTERS
2. HIST_EMERGENCY_VISITS
3. HIST_TOTAL_OOP

B. Clinical Burden Features

4. HIST_MEDICATION_COUNT
5. HIST_CONDITION_COUNT

C. Financial Feature

6. HIST_COVERAGE_RATE

D. Demographic Feature

7. AGE_2020

E. Social Determinants of Health

8. SDOH_COUNT

F. Encounter-Type Ratio Features

9. RATIO_ambulatory
10. RATIO_emergency
11. RATIO_inpatient
12. RATIO_outpatient
13. RATIO_urgentcare
14. RATIO_wellness

G. Top-10 Condition Binary Indicators (1 = had condition at least once in 2019–2020)

15. Hist_Condition_Stress (finding)
16. Hist_Condition_Part-time employment (finding)
17. Hist_Condition_Full-time employment (finding)
18. Hist_Condition_Not in labor force (finding)
19. Hist_Condition_Social isolation (finding)
20. Hist_Condition_Viral sinusitis (disorder)
21. Hist_Condition_Acute viral pharyngitis (disorder)
22. Hist_Condition_Acute bronchitis (disorder)
23. Hist_Condition_Limited social contact (finding)
24. Hist_Condition_Violence in the environment (finding)

H. Top-10 Medication Binary Indicators (1 = had medication at least once in 2019–2020)

25. Hist_Medication_Hydrochlorothiazide 25 MG Oral Tablet
26. Hist_Medication_Nitroglycerin 0.4 MG/ACTUAT Mucosal Spray
27. Hist_Medication_Simvastatin 10 MG Oral Tablet
28. Hist_Medication_Simvastatin 20 MG Oral Tablet
29. Hist_Medication_amLODIPine 2.5 MG Oral Tablet
30. Hist_Medication_insulin human isophane 70 UNT/ML / Regular Insulin Human

31. Hist_Medication_lisinopril 10 MG Oral Tablet
32. Hist_Medication_24 HR Metformin hydrochloride 500 MG Extended Release Oral Tablet
33. Hist_Medication_Clopidogrel 75 MG Oral Tablet
34. Hist_Medication_Amlodipine 5 MG Oral Tablet